

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		award (1) for any of following - phosphorus(III) chloride / phosphorus trichloride / PCl_3 - phosphorus(V) chloride / phosphorus pentachloride / PCl_5 - sulfur dichloride oxide / thionyl chloride / SOCl_2	1			1		1
		(ii)		71g chlorine in 239.1 g decanedioyl dichloride (1) 0.977g chlorine in $\frac{239.1}{71} \times 0.977 = 3.29$ g dichloride percentage purity = $\frac{3.29}{3.50} \times 100 = 94.0$ (1)		1				
		(iii)		moisture/water gained entry to the bottle and hydrolysed the decanedioyl dichloride when it was previously opened			1	1		1
				Question 12 total	4	3	9	16	2	2